Computational Physics Assignment Readme

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1 Problem Statement

A cook is tasked with cooking 20 pizzas at any one time using an oven. However, the cook does not bother whether a particular pizza is cooked properly. He randomly chooses one pizza and removes it from the oven, and replaces it with a new pizza. What is the distribution of cooking times for the pizza? The pizzas are replaced at regular intervals of 30 seconds and there is an infinite supply of pizzas. What is the distribution of cooking times of the pizzas? Write a program to simulate the cooking of the pizzas. The program only needs to obtain the cooking time for each pizza and get the distribution. Don't bother simulating the cooking process! Does the qualitative distribution change if there are 40 pizzas at one time?

2 Code

The code for this problem is written on MATLAB. The following image shows the code of the given problem

Computational_Physics_Assignment_Code.m

```
clear
clc
close all

q1 = input('Enter Number of Pizzas in the oven at a time -: ');
x = zeros(q1,1);        %q1 Pizzas are put inside the oven
q2 = input('Enter Total number of Pizzas outside the oven -: ');
a = zeros(q2,1);
for j = 1:q2            %Simulating the Random replacements of Pizzas and cooking times
    x = x+0.5;          %30 second increment
    r = randi([1 q1]); %Choosing a random pizza to take out
    a(j)=x(r);          %Labelling the cooking time for that pizza
    x(r)=0;             %Resetting the clock for that Pizza position
end
histogram(a,q2)         %Distribution plot
xlabel ('Time taken for the Pizza to get cooked (Minutes)')
ylabel ('Number of Pizzas cooked in the particular time interval')
```

Figure 1: MATLAB Code

3 Instructions to run the code

The following set of instructions can be for observing results from the code
1) Click on run in the editor tab of the command window
2) The first prompt will ask the user to enter the number of pizzas at a time. The problem statement states that at a time 20 pizzas are put at a time. However, there is also a test case where 40 pizzas are put inside the oven at a time. Hence, the different test cases can be analysed in the same code
3) The second prompt asks the user to decide as to how many pizzas are initially outside the oven. Preferably, this is a number of the order of 10^4 or 10^5 for more accurate results
4) The output of the code is a histogram of the number of pizzas cooked at particular time intervals

4 Test cases

The code can be verified using the following test cases

4.1 20 pizzas are inside the oven at a time and 10^6 pizzas are initially outside the oven

Here, the value taken by 'q1' is 20 and value taken by 'q2' is 1000000. This is the first test case that is given in the problem statement
The command window will be as follows

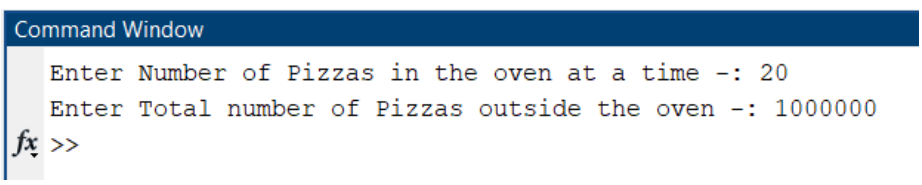

Figure 2: Command Window

The output will show the histogram of the distribution in this particular test case

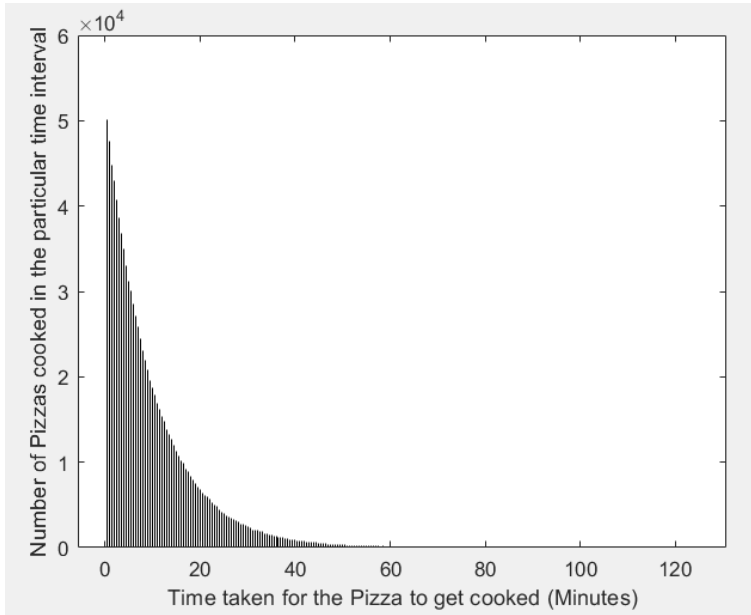

Figure 3: Distribution curve when 20 pizzas are inside the oven

4.2 40 pizzas are inside the oven at a time and 10^6 pizzas are initially outside the oven

Here, the value taken by 'q1' is 40 and value taken by 'q2' is 1000000. This is the second test case that is given in the problem statement
The command window will be as follows

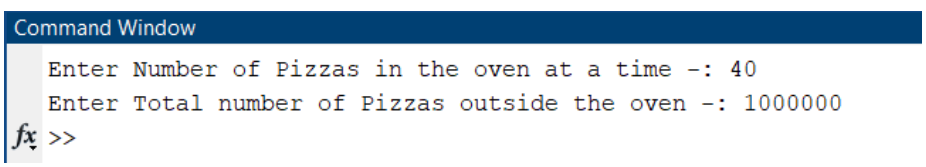

Figure 4: Command Window

The output is a distribution represented by the following histogram

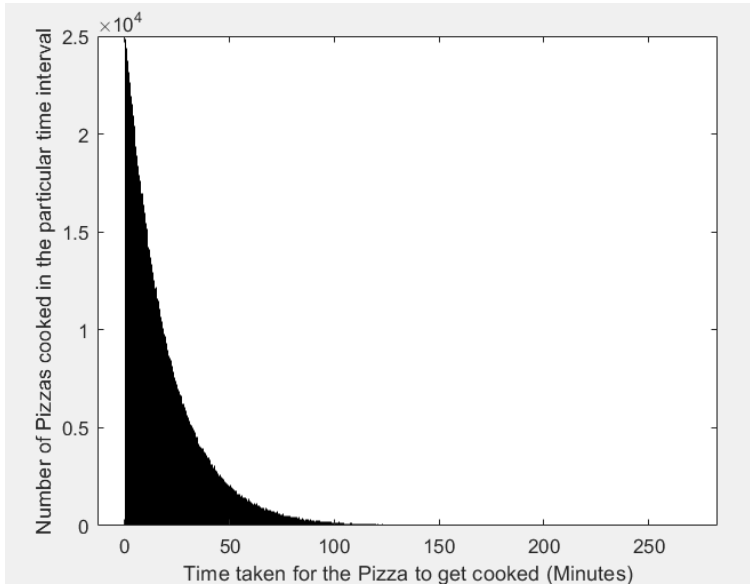

Figure 5: Distribution curve when 40 pizzas are inside the oven

4.3 100 pizzas are inside the oven at a time and 10^6 pizzas are initially outside the oven

Here, the value taken by 'q1' is 100 and value taken by 'q2' is 1000000. This is another test case that can be used to analyse the nature of the curve
The command window will be as follows

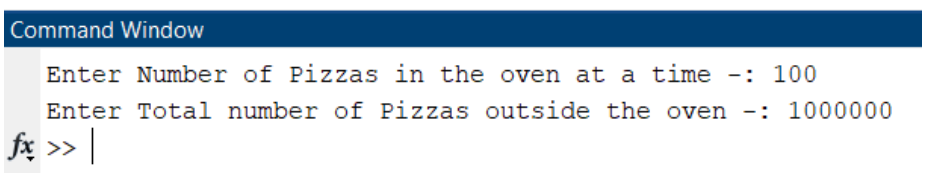

Figure 6: Command window

The output is a distribution represented by the following histogram

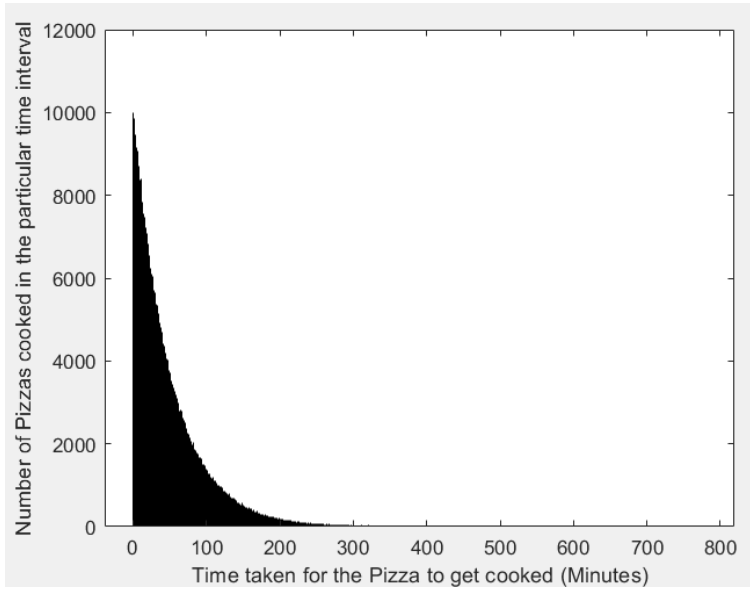

Figure 7: Distribution curve when 100 pizzas are inside the oven

4.4 1 pizza inside the oven at a time and 10^6 pizzas initially outside the oven

This is an edge case for testing the scenario where there is just one pizza at a time inside the oven. In this edge case, it is easily predictable that, at every iteration, the pizza that is in the oven has been cooked for 30 seconds. Then it is replaced by a raw pizza that also gets cooked for 30 seconds in the next iteration.The expected result is one line shooting up parallel to the Y axis at x = 0.5 (The 30 seconds mark). The command window is as follows

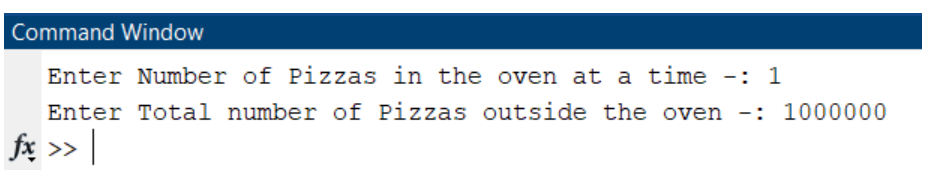

Figure 8: Command window

The distribution curve in this particular case is expressed as follows

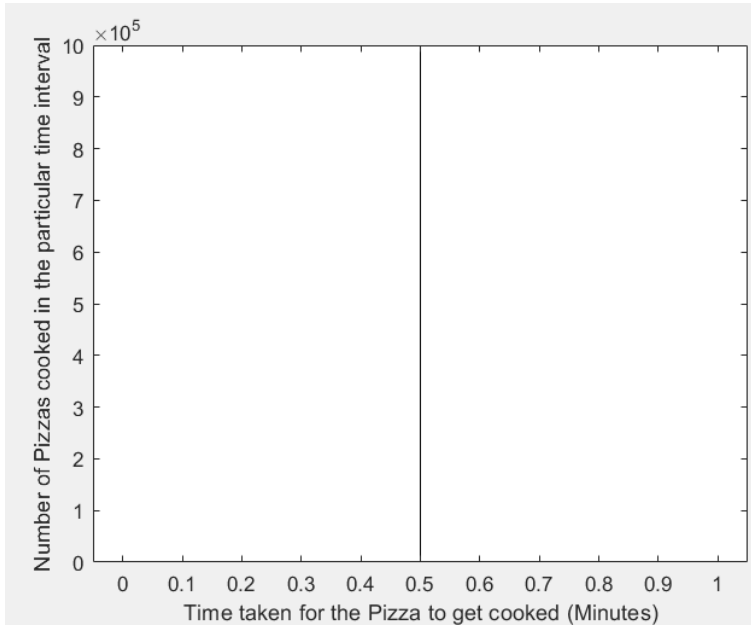

Figure 9: Distribution curve with 1 pizza inside the oven

The result acquired from the distribution curve is same as the same as the expected result. Hence, the code perfectly covers this particular edge case.

4.5 20 pizza inside the oven at a time and 1 pizza initially outside the oven

This is the second case for testing the scenario where there is just one pizza outside. The expected result is just one histogram bin from x = 0 to x = 1 of height q1 (In this case q1 = 1) since the readings of the pizzas that are taken outside are recorded. The command window is as follows

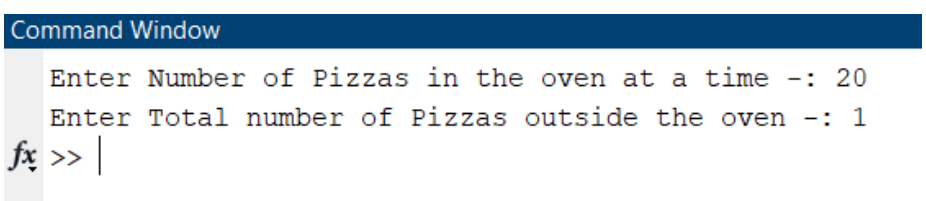

Figure 10: Command window

The distribution for this test case is

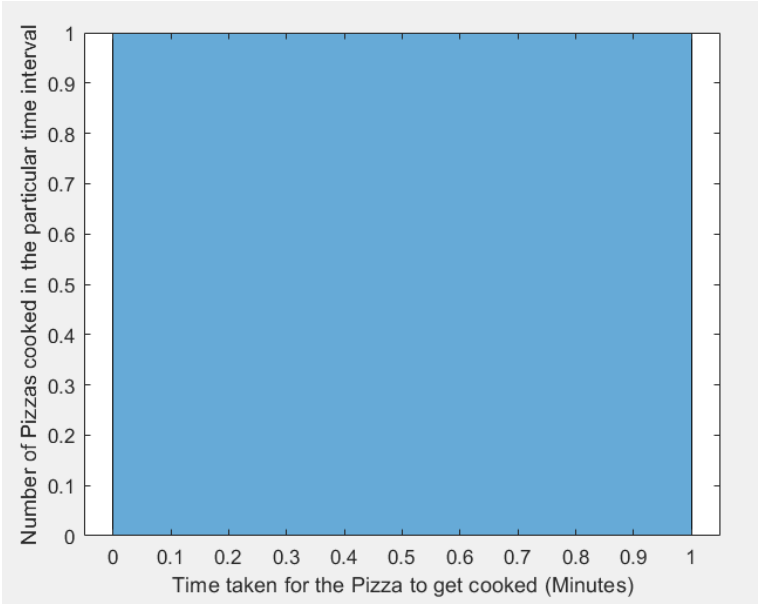

Figure 11: Distribution with just one pizza outside the oven